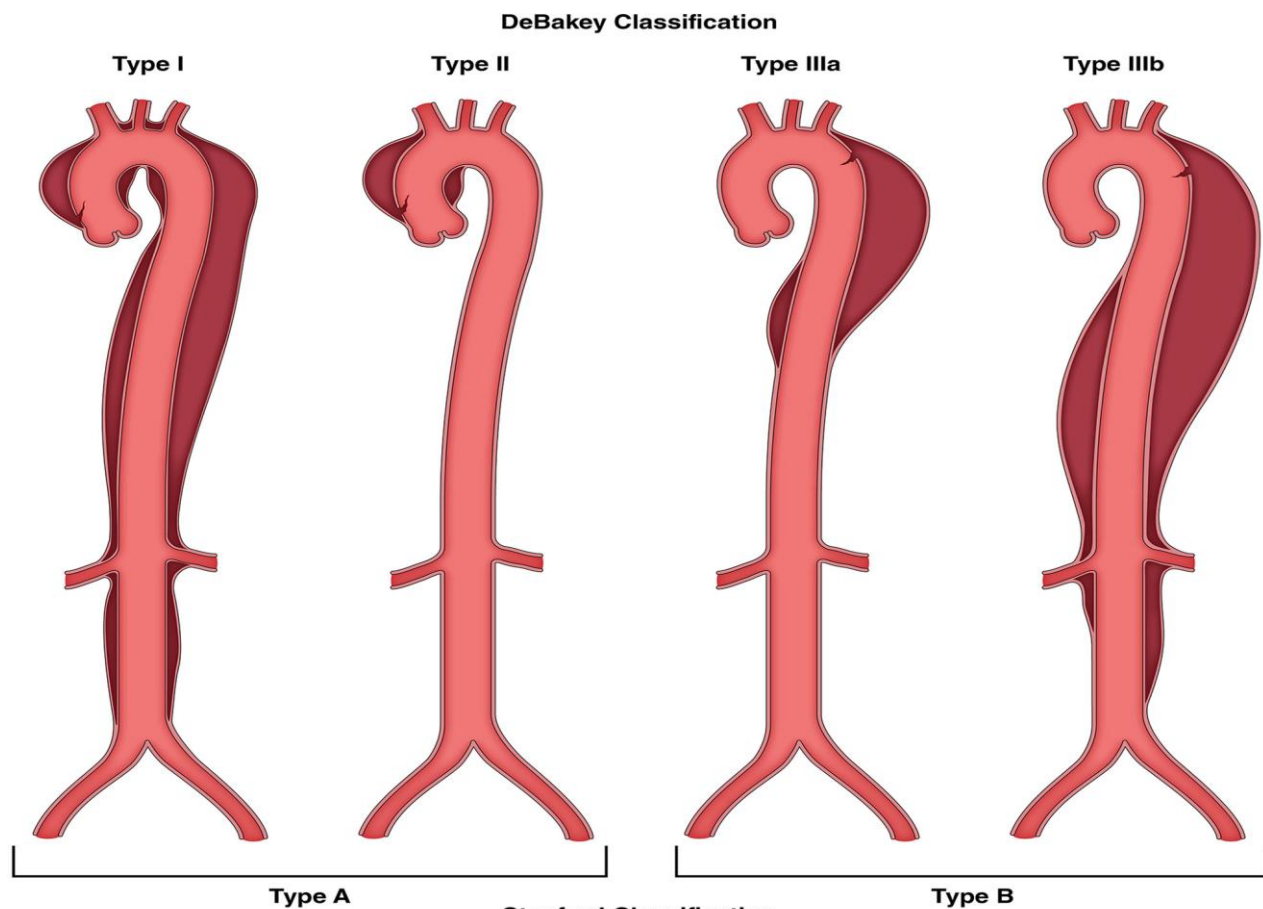


Objective: Aortic dissection (AD) is rare, life-threatening, and frequently difficult to recognize early because it can mimic more common emergency presentations. Our objective is to build an early-window prediction workflow for acute aortic dissection by focusing on patients with ECGs collected within the first 24 hours of ER/hospital presentation, preserving admission-level cohort structure. We used MIMIC-IV EHR and ECG data aiming to develop a model capable of outperforming current clinical misdiagnosis rates in acute AD events.

Background:

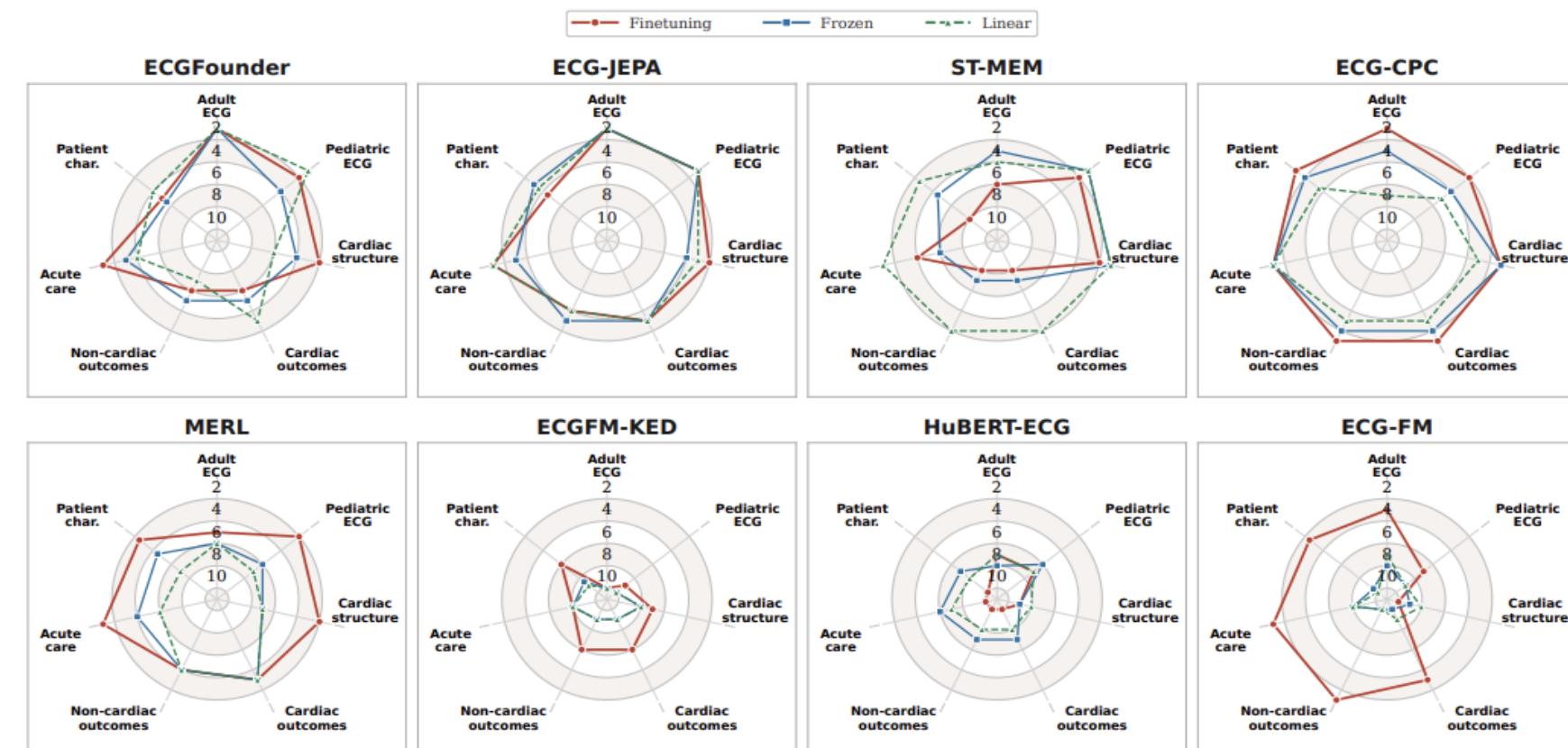
Aortic dissection occurs when a tear in the inner wall of the aorta allows blood to split vessel wall layers, creating a false lumen. It is rare, but can be fatal when missed, and there is a large overlap in presentation with more common conditions like myocardial infarction or pulmonary embolism.



Stanford Type A dissections, classified by involvement of the ascending aorta, have poor prognosis unless treated immediately by surgical repair, with mortality increasing 1-2% each hour after onset of symptom. Stanford Type B dissections, involving the descending aorta, are more manageable, requiring aggressive blood pressure and heart rate control.

Definitive diagnosis begins with clinical suspicion (sudden back and chest pain) and requires CT angiography; however most emergent cases only initially receive Electrocardiograms (ECGs) and standard lab tests, leading to approximately 1/3 of acute aortic dissections being misdiagnosed. An ECG is not reliable for diagnosing or excluding acute aortic dissection, while they can potentially contain warning signals that are not fully used in routine assessment (Xie et al. 2020).

Foundation Models (FM) are trained initially on a set of data that provides general structure, and are subsequently applied to a small, highly specific dataset. Typically, this leads to substantial improvement in model performance. Al-Masud et al. (2026) evaluated the performance of several ECG-FM models across different backbone and pretraining types and sample sizes:



However, previous studies on ML's potential in diagnosing AD specifically have been largely heterogenous, small-scale, and with mixed results, and haven't used FMs (Mau et al. 2026). Our objective is to first build an XGBoost model to assess our data's capabilities and subsequently train an FM to improve AD predictive power.

Methods:

- Dataset: The publicly available MIMIC-IV dataset, containing records of real patient visits from a medical center in Boston, was used as the sole source of EHR and ECG records.
- Preprocessing: Patients under 18 were removed from the study. Patients were filtered to ensure at least one ECG reading within 24 hours of their initial hospital encounter, and an attached discharge note.
- LLM-driven parsing using the OpenAI library (model GPT-5.6-luna) was performed on discharge notes, identifying targets by chief complaint and imaging or summary notes. Stanford Type B dissections were removed from analysis.
- Interestingly, a large discrepancy between LLM confirmations and ICD-coded encounters was found.
- 102 patients were labeled as having ADs but lacked ICD codes. These were sometimes due to chronic AD or the LLM misinterpreting a history of AD, but some human error was noted.
- 110 patients had ICD codes but lacked an LLM-supported confirmation. These appeared to be mostly human error.
- Controls were selected from non-target admissions with chest/back pain phenotypes and LLM-supported usefulness as plausible AD mimics or non-aortic-syndrome encounters
- ECG and lab measurements were only considered within 24 hours of the initial encounter time. Demographics are taken from the encounter as opposed to historical averages. QT and Bazett-corrected QTc intervals were derived.
- XGBoost was initially used to determine a baseline, with a standard 80/20 split. Model performance was evaluated using PR-AUC, ROC-AUC, and SHAP summaries.
- An FM was trained on 800K MIMIC-IV ECGs, before transferring to the AD cohort.

Results:

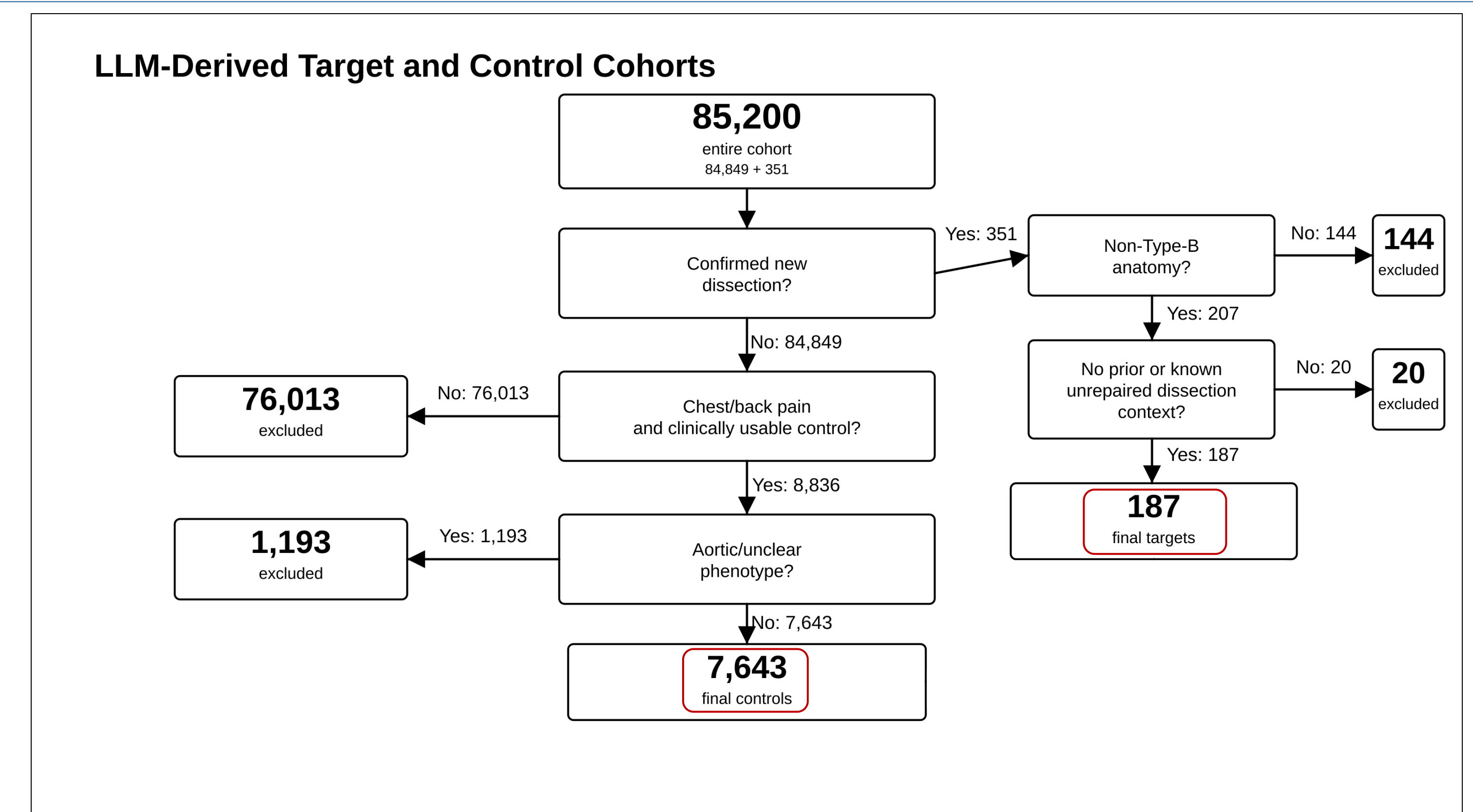


Fig. 1: Target and control cohort derivation flowchart. Phenotypes of the groups are determined using LLM parsing of discharge notes.

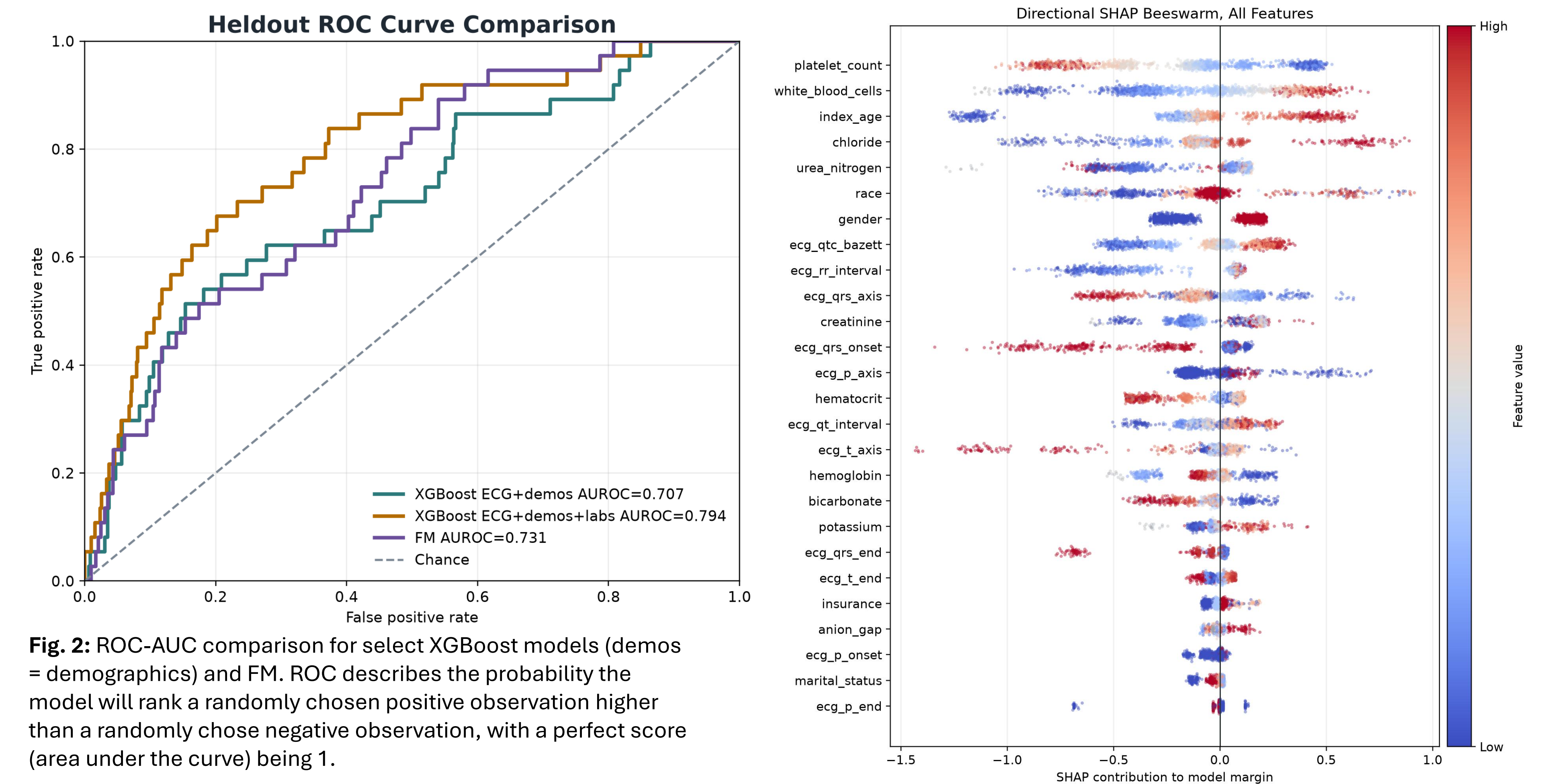


Fig. 2: ROC-AUC comparison for select XGBoost models (demos = demographics) and FM. ROC describes the probability the model will rank a randomly chosen positive observation higher than a randomly chose negative observation, with a perfect score (area under the curve) being 1.

The FM (0.731) outperforms XGBoost without labs (0.707), but underperforms XGBoost with labs (0.794).

Figure 3: XGBoost SHAP Beeswarm plot, feature set ECGs + demographics + labs. A SHAP Beeswarm plot colors a feature's datapoint based on its literal value, and places it left-to-right based on its importance in the model's prediction. The features most influential are placed at the top.

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Summary:

- Final cohort sizes were 7643 controls, and 187 cases. The holdout ROC-AUC is strongest for the XGBoost model with labs, ECGs, and demographics, at 0.794. This is much higher than the ROC-AUC for the other models, and demonstrates the model learns many of the patterns within the data, especially given the small number of cases.
- Models that incorporated laboratory features performed better, but these results carry greater leakage risk because early labs, vitals, and missingness patterns may reflect clinician suspicion or downstream diagnostic workup. Platelet count and white blood cell count were the most important features, consistent with physiologic stress and inflammatory pathways.
- The FM performed well, earning an ROC-AUC of 0.731, despite a more limited feature set than the other models.
- This study demonstrates a measurable but limited early EHR signal for acute AD. Performance constraints were pervasive: small number of cases due to relative rarity of AD; high misdiagnosis rate of the condition in general; inability to determine exact time of dissection, which was often before admission to the ED; XGBoost handling of missingness forced removal of non-prevalent features; etc. It should also be noted that ECGs are supplementary to official diagnosis protocol, not primary.

Conclusion:

- FMs can improve acute AD prediction, but modelling is overall limited by phenotypic uncertainty and a small number of cases.
- LLM-based discharge-note phenotyping provides a cleaner cohort definition than using ICD codes alone.
- Future work will benefit from the inclusion of LLM-parsed radiology notes of other formats, enlargement of the target cohort, and inclusion of lab measurements in the FM training set.